

Supplementary information, methods, and figures accompanying Goltz and Krawczynski (2026) *Volcanica*

Supplementary information: Preparation of starting mixes for EPMA analysis and potential sources of heterogeneity

Synthetic powders were melted for a short (on the order of minutes to hours) duration above the liquidus temperature and quenched in a one-atmosphere furnace. Glass beads were then mounted in epoxy and prepared for EPMA analysis. The resulting glass was analyzed with electron microprobe with a broad beam, and the standard deviation (not standard error) is reported. Short durations in the 1-atm furnace used for this 'glassing' step result in higher standard deviations due to small amounts of heterogeneity, however, this prevents volatilization of alkalis and iron. We thus feel this gives the best evaluation of the actual starting material composition. Experiments are run for much longer durations, and it is shown that the electron microprobe analyses of the experimental glasses are far more homogenous and equilibrated; thus heterogeneity in the starting material is unlikely to impact the results of our study substantially, if at all.

Supplementary information: calibration of piston cylinder apparatus

Pressure and temperature calibrations of both piston cylinder apparatuses were performed using identical ½" cell assemblies. First, the pressure was calibrated at room temperature using the resistivity transition of bismuth (^{1,2}). At high temperature, the pressure calibration was investigated using the kushiroite (CaAlAlSiO₆) breakdown at low pressure into a mixture of anorthite, gehlenite and corundum and its high-pressure phase transition into grossular and corundum (³). Results show that no pressure correction is needed from the calculated values due to factors like internal friction or assembly strength. Temperature calibration and thermal gradients inside the capsule were extensively studied using 2 different methods. The first method uses two thermocouples with one sitting centered axially at the position of a would-be sample in the assembly, and the other sitting at its standard reading position. The temperature difference gives an offset of +40°C between the reading thermocouple and the capsule center. A spinel growth thermometer was used as the secondary method (⁴). It confirms the 40°C offset between thermocouple and capsule center and maps out the temperature gradient across the capsule.

As for the ¾" cell assembly, the pressure was also calibrated using the breakdown of kushiroite phase transition. It shows that no pressure correction is necessary. Temperature calibration using the 2-thermocouple technique shows that the temperature offset is

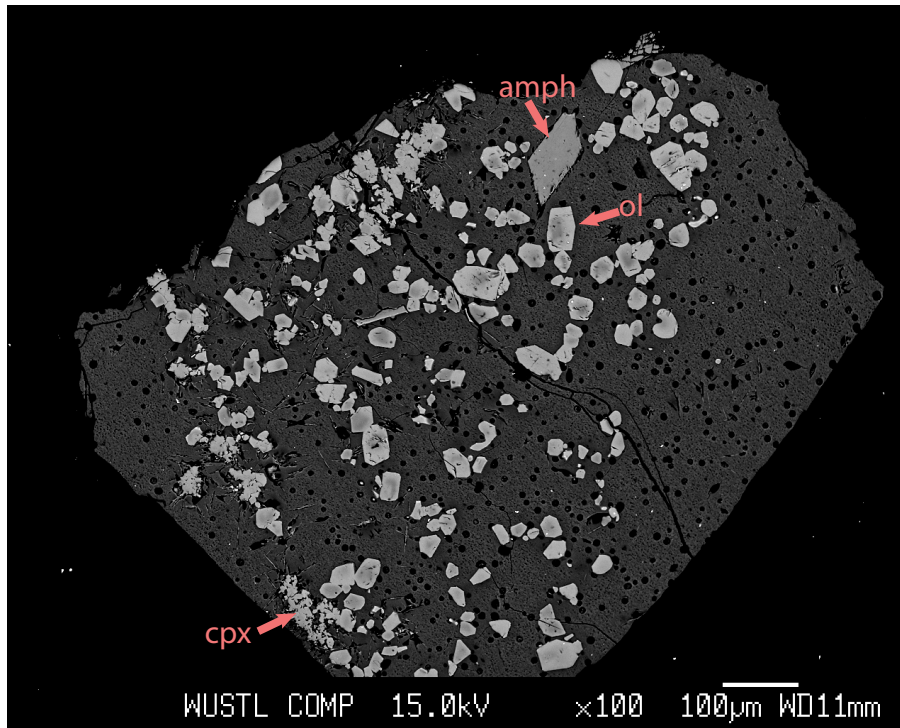
pressure dependent and varies linearly from 31°C at 0.5 GPa to 21°C at 1 GPa. Temperature was controlled using a Eurotherm PID controller and type C thermocouple.

We estimate a 15°C temperature uncertainty for both assemblies.

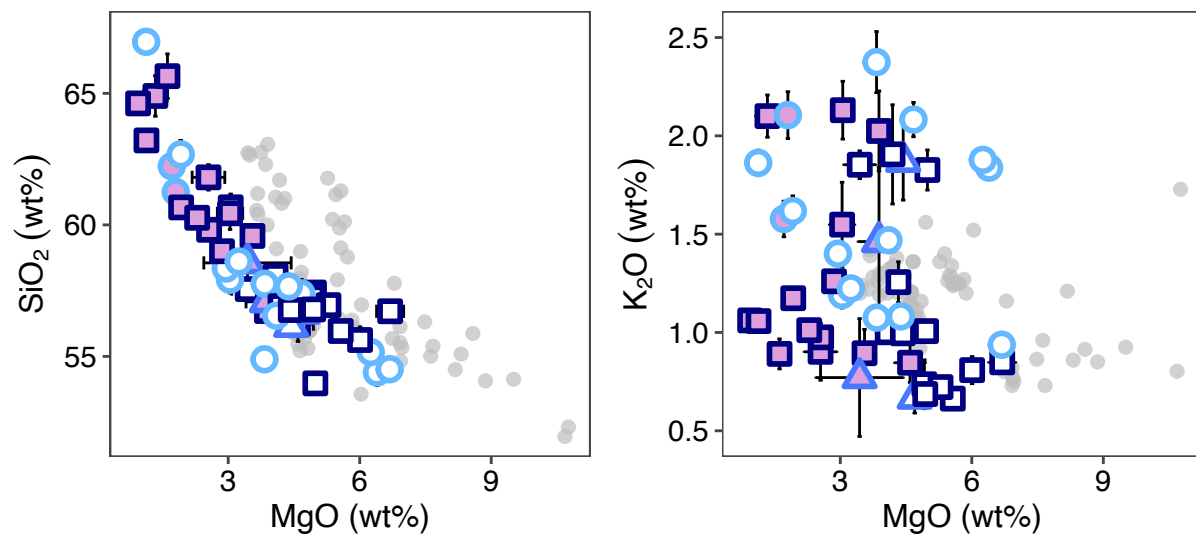
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- ⁴⁾ E.B. Watson, D. A. Wark, J.D. Price, J.A. Van Orman, "Mapping the thermal structure of solid-media pressure assemblies" *Contributions* 142: 640-652 (2002)
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Supplementary methods: Analysis of natural amphibole crystals at Carnegie Science via EPMA

Data was collected on a JEOL 8530F electron microprobe equipped with 5 wavelength-dispersive spectrometers. Operating conditions were an accelerating voltage of 15kV, a beam current of 20nA, and a beam diameter of 3 microns. Data was collected using the Probe for EPMA software package. Count times were 20 seconds (on peak) and 10 seconds (upper and lower background), with the exception of the volatile elements Na and K (10 seconds on peak and 5 seconds per background measurement). The standards used were Ilmenite for Ti ka, Mn ka; Anorthite for Al ka; Ca ka; Albite for Na ka; MgCr₂O₄ for Cr ka; Orthoclase for K ka; Springwater Olivine for Mg ka, Si ka; and Rockport Fayalite - for Fe ka. Na, Mg, Si, and Al were analyzed using a TAPL crystal, Ti and Cr using a PETL crystal, Fe and Mn on a LIFL crystal, and K and Ca on a PETH crystal.



Supplementary Figure 1. Image of OD102 demonstrating core-rim zoning in Fe-Mg of olivine crystals.



Supplementary Figure 2. Harker diagram showing variability in SiO₂ and K₂O in our experiments as compared to natural samples. Colors and shapes are as in Figure 6.